LAB 6

TASK 1

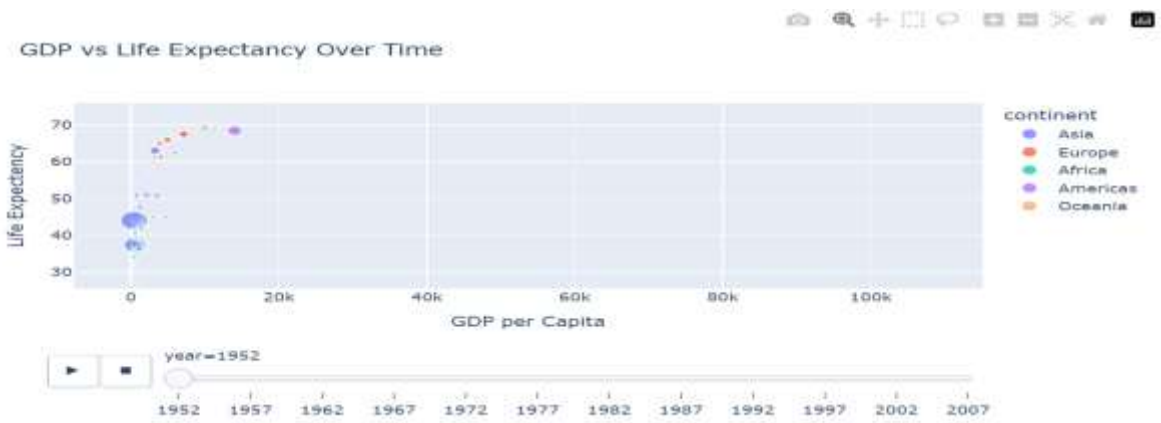
Scatter Plot with Slider (Animation)

- Plot GDP per capita (gdpPercap) vs life expectancy (lifeExp) for all countries.
- Use population (pop) to size the points.
- Color points by continent.
- Add a slider to move through different years.
- Title: "GDP vs Life Expectancy Over Time".
- Label the x-axis as "GDP per Capita (USD)" and the y-axis as "Life Expectancy (Years)".
- Ensure hover tooltips display country names.

INPUT

```
import plotly.express as px

data = px.data.gapminder()

data.head()


fig = px.scatter(data, x="gdpPercap", y="lifeExp", title='GDP vs Life Expectancy Over Time',size="pop", labels={'gdpPercap': "GDP per Capita", 'lifeExp': "Life Expectency"}, animation_frame='year', hover_name='country', color='continent')

fig.show()
```

OUTPUT



TASK 2

Line Plot with Dropdown Selector

- Plot life expectancy (lifeExp) vs year (year) for multiple countries (e.g., India, China, USA, Brazil).

- Add a dropdown menu to select which country's line is visible.

- Use distinct line colors for each country.

- Title: "Life Expectancy Comparison by Country".

- Label the x-axis as "Year" and the y-axis as "Life Expectancy (Years)".

- Place the legend at the bottom.

INPUT

OUTPUT

TASK 3

Bar Plot with Buttons (Switch Between Variables)

- Plot population (pop) by continent for 2007.

- Add buttons to toggle between population and GDP per capita (gdpPercap).

- Use grouped bars for clarity.

- Title: "Continent Comparison (2007)".

- Label the x-axis as "Continent" and the y-axis as "Value".

- Display the legend at the top-right corner.

INPUT

OUTPUT

TASK 4

Map with Projection Selector

- Plot life expectancy (lifeExp) by country on a world map for 2007.

- Use population (pop) to size the markers.

- Add hover tooltips to show country names.

- Add buttons to switch between map projections (Natural Earth, Mercator, Orthographic).

- Title: "World Map of Life Expectancy (2007)".

INPUT

OUTPUT

TASK 5

Animated Scatter Plot with Play/Pause Controls

- Create an animated scatter plot of GDP per capita vs life expectancy across years.

- Use population (pop) for point size.

- Color points by continent.

- Add play/pause controls to animate automatically.

- Title: "Global Development Trends (1952–2007)".

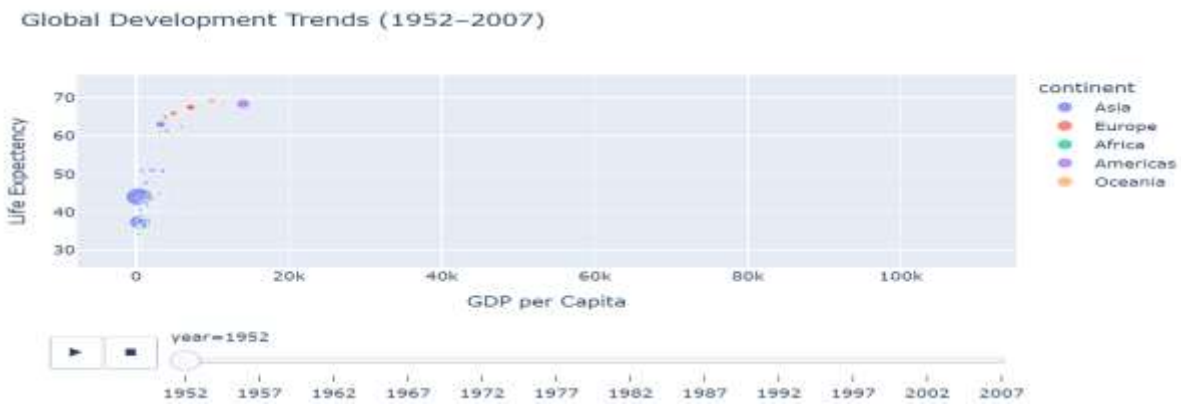
- Label axes appropriately.

INPUT

fig = px.scatter(data, x="gdpPercap", y="lifeExp", title='Global Development Trends (1952–2007)',size="pop", labels={'gdpPercap': "GDP per Capita", 'lifeExp': "Life Expectency"}, animation_frame='year', hover_name='country', color='continent')

fig.show()

OUTPUT



TASK 6

3D Scatter Plot

- Plot GDP per capita (gdpPercap), life expectancy (lifeExp), and population (pop) in 2007.

- Use continent for color coding.

- Size points by population.

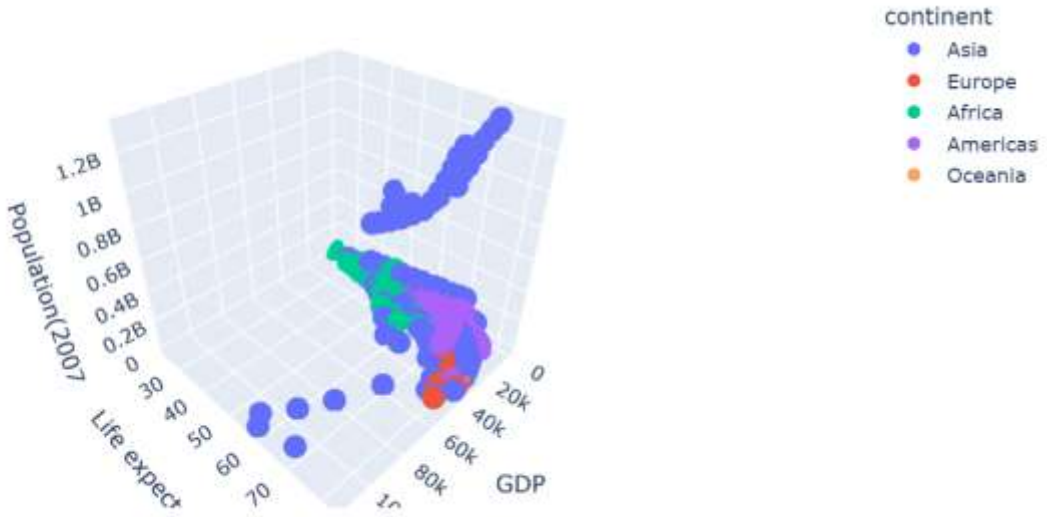
- Title: "3D Scatter: GDP, Life Expectancy, Population (2007)".

- Label axes appropriately.

INPUT

```
fig = px.scatter_3d(data, x='gdpPercap', y='lifeExp', z='pop', color='continent',
labels={'gdpPercap': 'GDP', 'lifeExp': "Life expectency", 'pop' : "Population(2007"})

fig.show()
```

OUTPUT



TASK 7

Heatmap

- Compute the correlation matrix for lifeExp, gdpPercap, and pop.

- Plot the matrix as a heatmap.

- Add numeric values inside each cell.

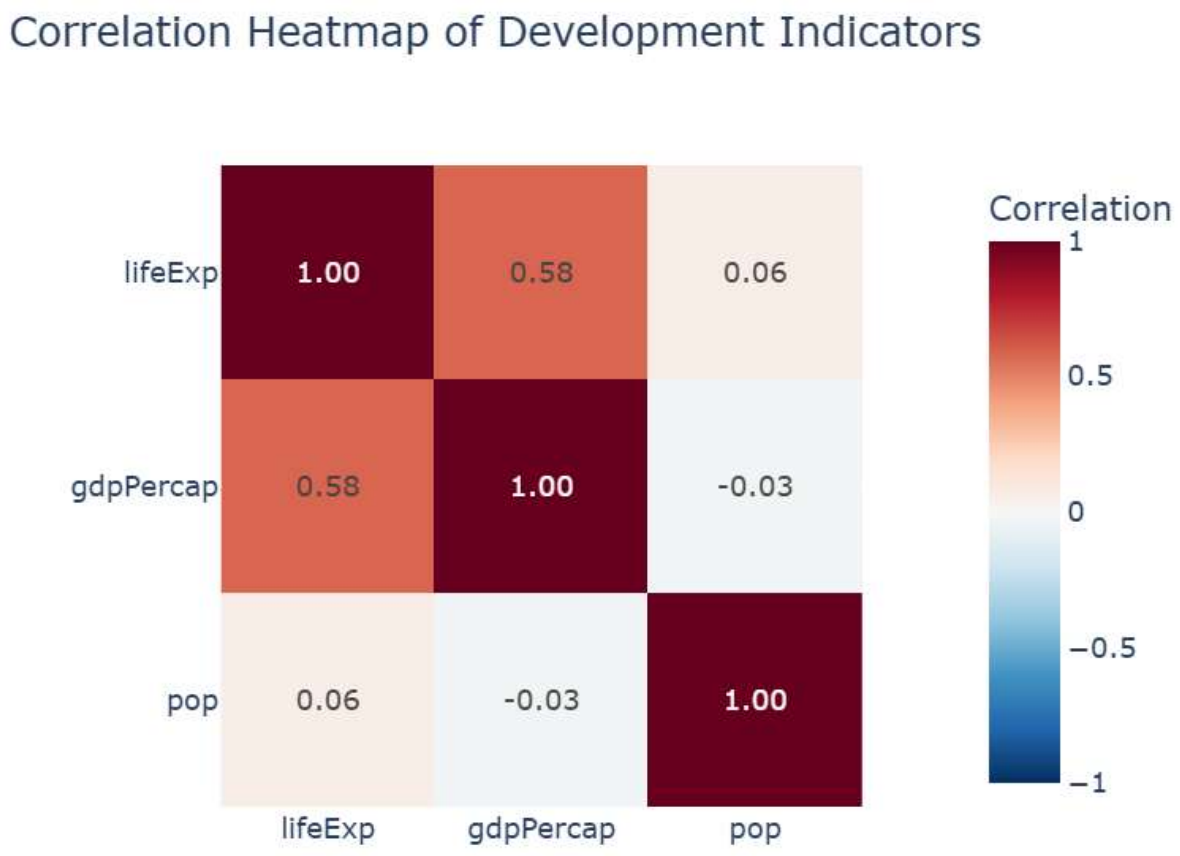
- Title: "Correlation Heatmap of Development Indicators".

- Use a diverging color scale for better contrast.

INPUT

```
corr_matrix = data[["lifeExp", "gdpPercap", "pop"]].corr()


fig = px.imshow( corr_matrix,   text_auto=".2f",
        color_continuous_scale="RdBu_r", zmin=-1,
        zmax=1,
        title="Correlation Heatmap of Development Indicators",
        labels={"color": "Correlation"})


fig.show()
```

OUTPUT



TASK 8

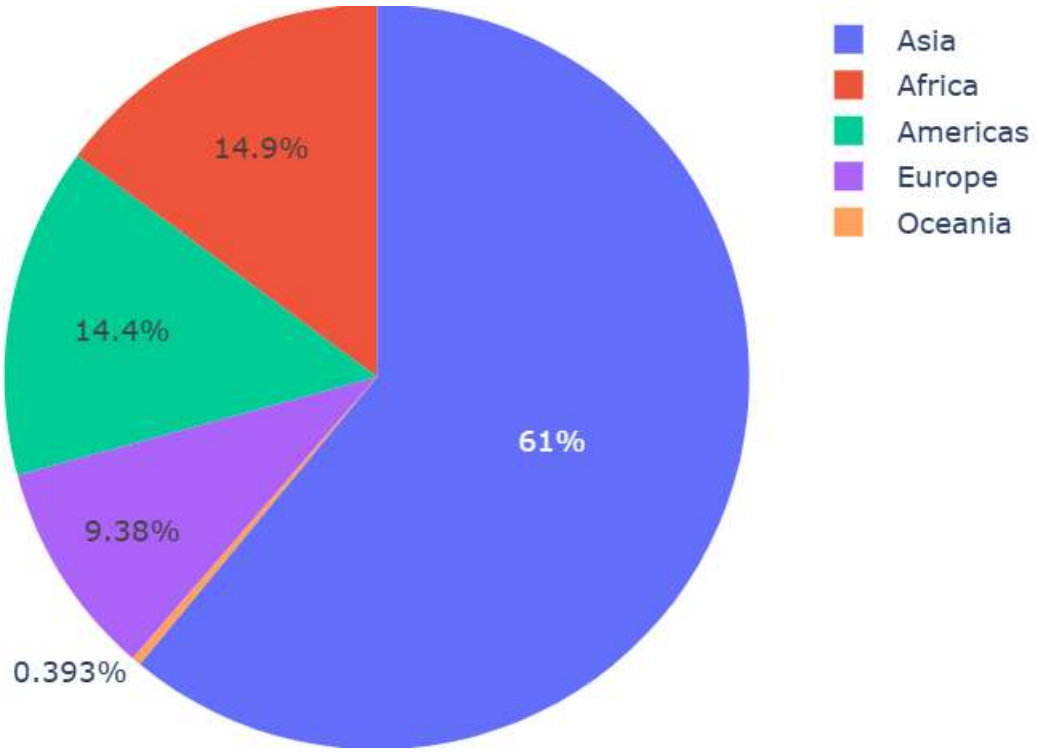
Pie Chart

- Create a pie chart showing population share (pop) by continent in 2007.
- Add percentage labels to each slice.
- Title: "Population Share by Continent (2007)".
- Use contrasting colors for slices.

INPUT

```
data_2007 = data[data.year==2007]


pie_plot = px.pie(
   data_2007,
   names="continent",
   values="pop"
)


pie_plot.show()
```

OUTPUT



TASK 9

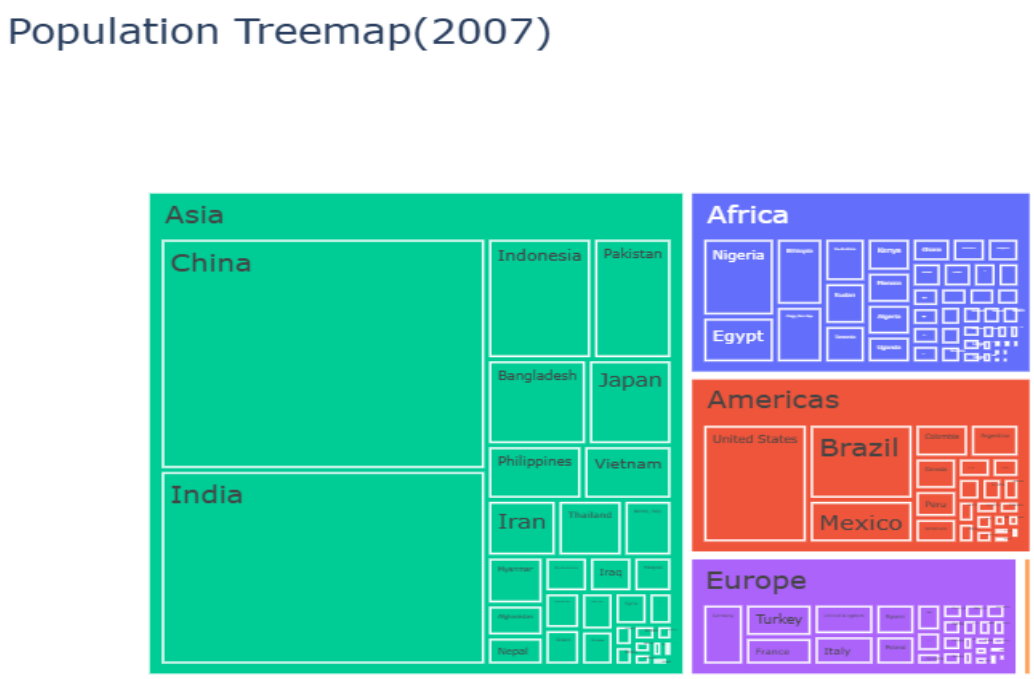
Treemap of Population

- Title: "Population Treemap (2007)"
- Size blocks by population.
- Color blocks by continent.
- Hover tooltips should show country names.

INPUT

```
treemap_plot = px.treemap(
    data_2007,
    path=["continent", "country"],
    values="pop",
    color="continent",
    hover_name="country",
    title="Population Treemap(2007)"
)
treemap_plot.show()
```

OUTPUT



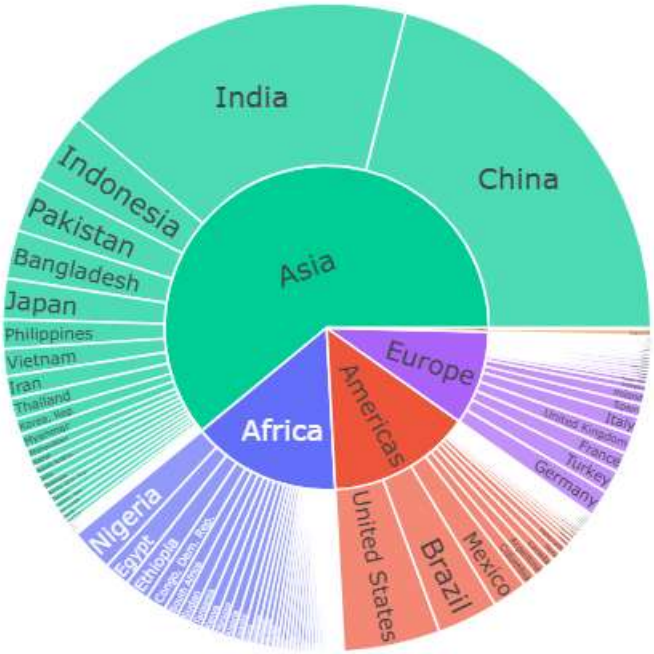
TASK 10

Sunburst Chart

- Title: "Sunburst of Population (2007)".
- Inner ring = continent, outer ring = country.

INPUT

```
sunburst_plot = px.sunburst(
    data_2007,
    path=["continent", "country"],
    values="pop",
    color="continent",
    hover_name="country",
    title="Subburst of Population(2007)"
)
sunburst_plot.show()
```

OUTPUT

Sunburst of Population(2007)